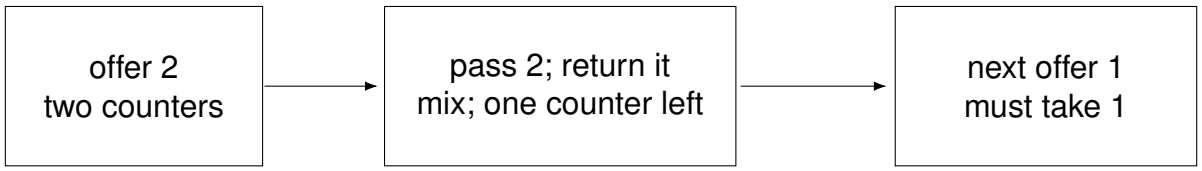
Each bag has one ticket marked 0, one marked 4 and one marked 6. Each has the same chance. Return the ticket and mix after *every* draw. The next offer is a fresh draw from the same bag.

Set the number of offers before a round. Your partner shows each offer. Take it to end the round, or pass it forever. With one counter left, you must take the offer, even if it is 0. Remove one turn counter after deciding on each offer. Only the offer you take scores.

A practice round uses different tickets: 1, 2, 5.



round score: 1

Problem 1: Use the 0, 4, 6 bag. Invent a taking rule for two-offer rounds and one for three-offer rounds. Keep each rule fixed while playing six rounds with a partner. Try to get a high total score. Swap roles.

Two-offer rule _____

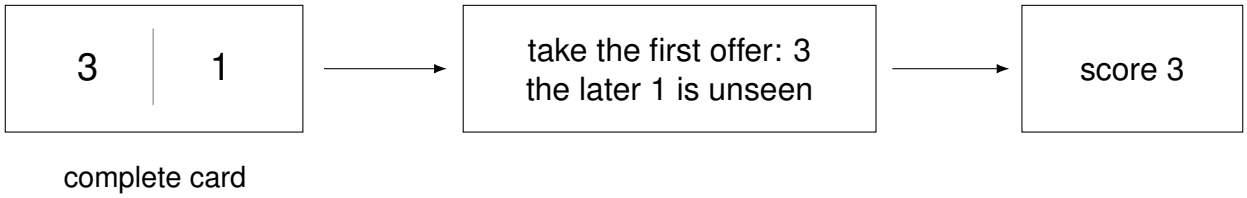
Three-offer rule _____

Round	1	2	3	4	5	6	Total
Two offers							
Three offers							

Problem 2: Can the same fixed rules give opposite winners in two six-round matches? Show possible offers for both. Can either match settle which rule has the better average over many rounds?

A card lists offers in drawing order, including offers that would remain unseen after you take one. All complete cards of the same length have the same chance.

With practice tickets 1, 3, 5, this rule takes 3 or 5 and passes 1:



Problem 3: Find first-offer choices for 0, 4 and 6 that give the highest total score on all nine cards. Use the same choices on every card. The second offer must be taken if you reach it.

0 | 0 score _____	0 | 4 score _____	0 | 6 score _____
4 | 0 score _____	4 | 4 score _____	4 | 6 score _____
6 | 0 score _____	6 | 4 score _____	6 | 6 score _____

First offer 0: _____

First offer 4: _____

First offer 6: _____

Total score on nine cards: _____

Problem 4: Which plan gives the higher total score on these 27 three-offer cards? Use every card for each plan, including cards with the same score.

A: Take 4 or 6 whenever it appears. Pass 0.

B: Take only 6 on the first offer. On the second, take 4 or 6 and pass 0.

Both plans take the last offer if they reach it.

A's total: _____ B's total: _____

0 | 0 | 0 score _____	0 | 0 | 4 score _____	0 | 0 | 6 score _____
0 | 4 | 0 score _____	0 | 4 | 4 score _____	0 | 4 | 6 score _____
0 | 6 | 0 score _____	0 | 6 | 4 score _____	0 | 6 | 6 score _____
4 | 0 | 0 score _____	4 | 0 | 4 score _____	4 | 0 | 6 score _____
4 | 4 | 0 score _____	4 | 4 | 4 score _____	4 | 4 | 6 score _____
4 | 6 | 0 score _____	4 | 6 | 4 score _____	4 | 6 | 6 score _____
6 | 0 | 0 score _____	6 | 0 | 4 score _____	6 | 0 | 6 score _____
6 | 4 | 0 score _____	6 | 4 | 4 score _____	6 | 4 | 6 score _____
6 | 6 | 0 score _____	6 | 6 | 4 score _____	6 | 6 | 6 score _____

Problem 7: For each new bag, find all the first-offer choices that give the highest total score on its nine two-offer cards. Each bag has one of each shown ticket; return and mix after every draw.

Bag: 0, 3, 6

0 | 0 score _____	0 | 3 score _____	0 | 6 score _____
3 | 0 score _____	3 | 3 score _____	3 | 6 score _____
6 | 0 score _____	6 | 3 score _____	6 | 6 score _____

Choices: _____ Total: _____

Bag: 0, 5, 6

0 | 0 score _____	0 | 5 score _____	0 | 6 score _____
5 | 0 score _____	5 | 5 score _____	5 | 6 score _____
6 | 0 score _____	6 | 5 score _____	6 | 6 score _____

Choices: _____ Total: _____